In [1]:
```
import matplotlib.pyplot as plt
```

In [2]:
```
import numpy as np
```

Line Chart

In [5]:
```
x = range(101,111)
```

In [7]:
```
y_PDS = np.random.randint(35,101,10)
```

In [9]:
```
y_PDS
```

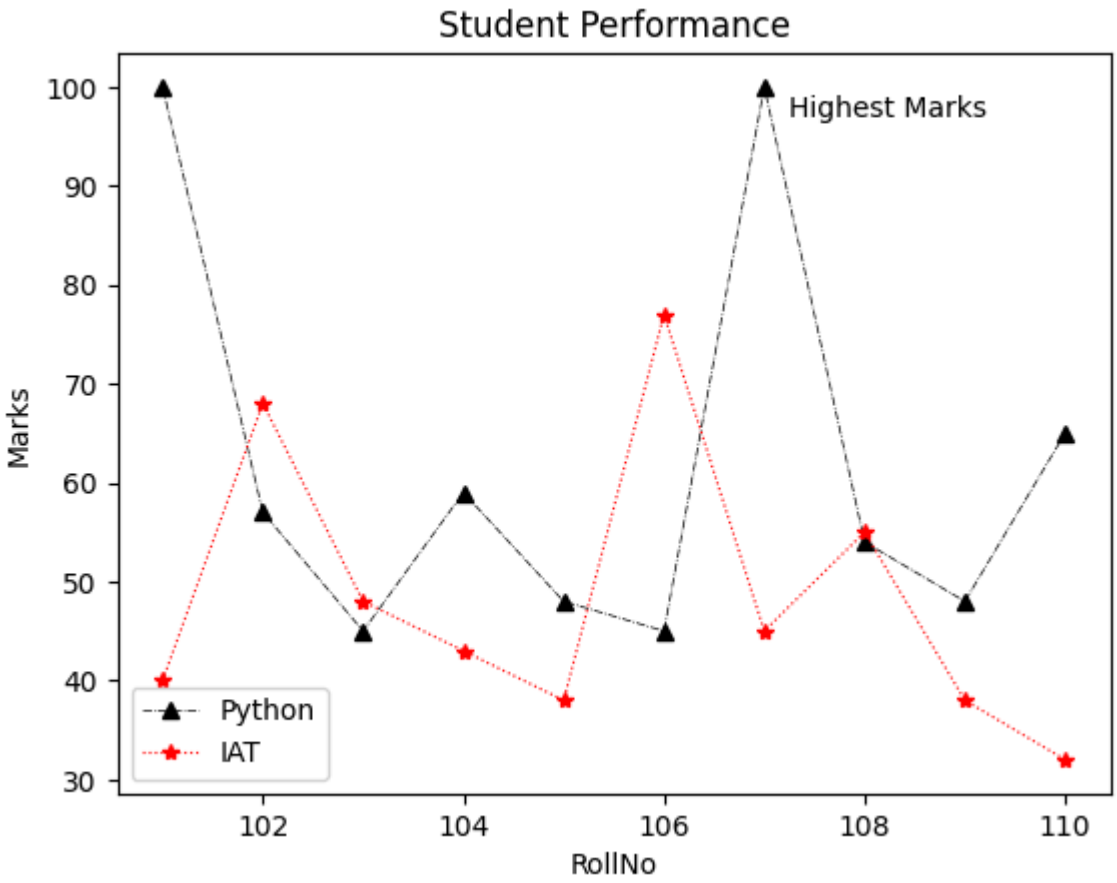
Out[9]:
```
array([100,  57,  45,  59,  48,  45, 100,  54,  48,  65], dtype=int32)
```

In [11]:
```
y_DM = np.random.randint(30,85,10)
```

In [13]:
```
y_DM
```

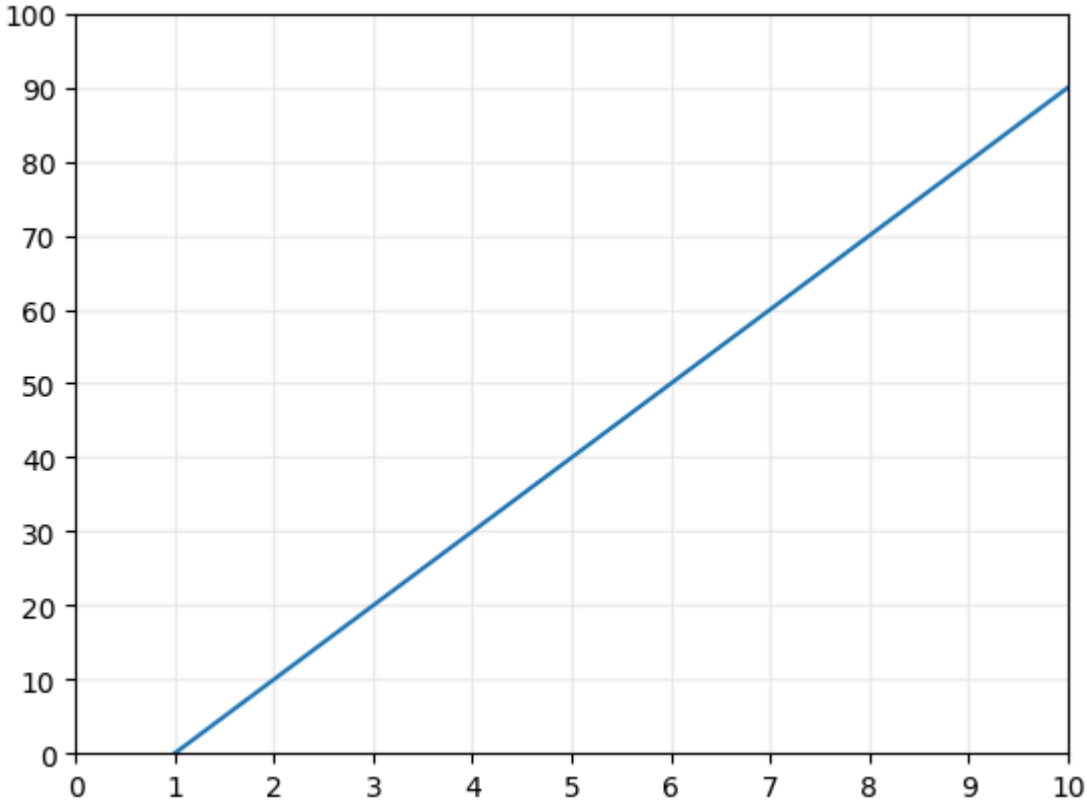
Out[13]:
```
array([40, 68, 48, 43, 38, 77, 45, 55, 38, 32], dtype=int32)
```

In [15]:
```
plt.plot(x, y_PDS, marker = "^", linestyle ="-.", lw = 0.5, c = "k")
plt.plot(x, y_DM, marker = "*", ls = ":", linewidth = 0.8, c = "r")
plt.xlabel("RollNo")
plt.ylabel("Marks")
plt.title("Student Performance")
plt.legend(["Python","IAT"])
plt.annotate(xy = [107.25,97], text = "Highest Marks")
plt.savefig("MCA_3_Marks.png")
plt.show()
```



Custom Axis

In [17]:
```
ax = plt.axes()
ax.set_xlim(0,10)
ax.set_ylim(0,100)
ax.set_xticks(range(0,11))
ax.set_yticks(range(0,101,10))
ax.grid(alpha = 0.2)
plt.plot(range(1,11), range(0,100,10))
plt.show()
```

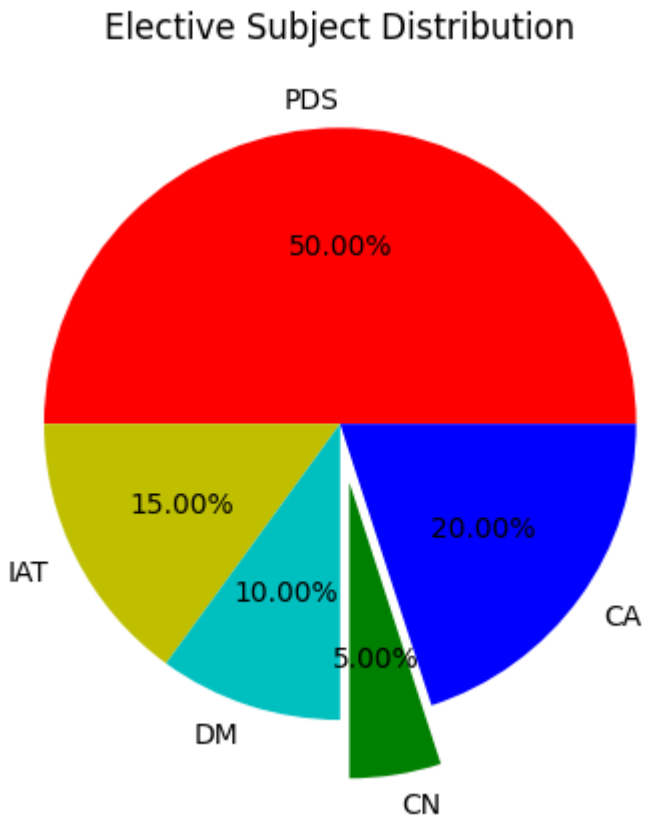


Pie Chart

In [19]:
```
student_count = [50,15,10,5,20]
```

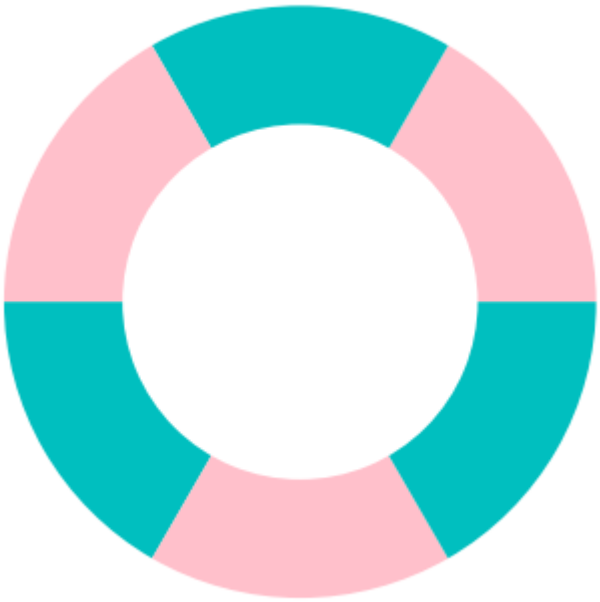
In [21]:
```
subject = ["PDS","IAT","DM","CN","CA"]
```

In [23]:
```
plt.pie(student_count, labels=subject, colors=["r","y","c","g","b"], radius = 1,
        explode = [0,0,0,0.2,0])
plt.title("Elective Subject Distribution")
plt.show()
```



Donut Chart

In [26]:
```
plt.pie([10,10,10,10,10,10], radius = 1, colors = ["pink","c","pink","c","pink",
plt.pie([45], colors = ["w"], radius = 0.6)
plt.show()
```



Bar Chart

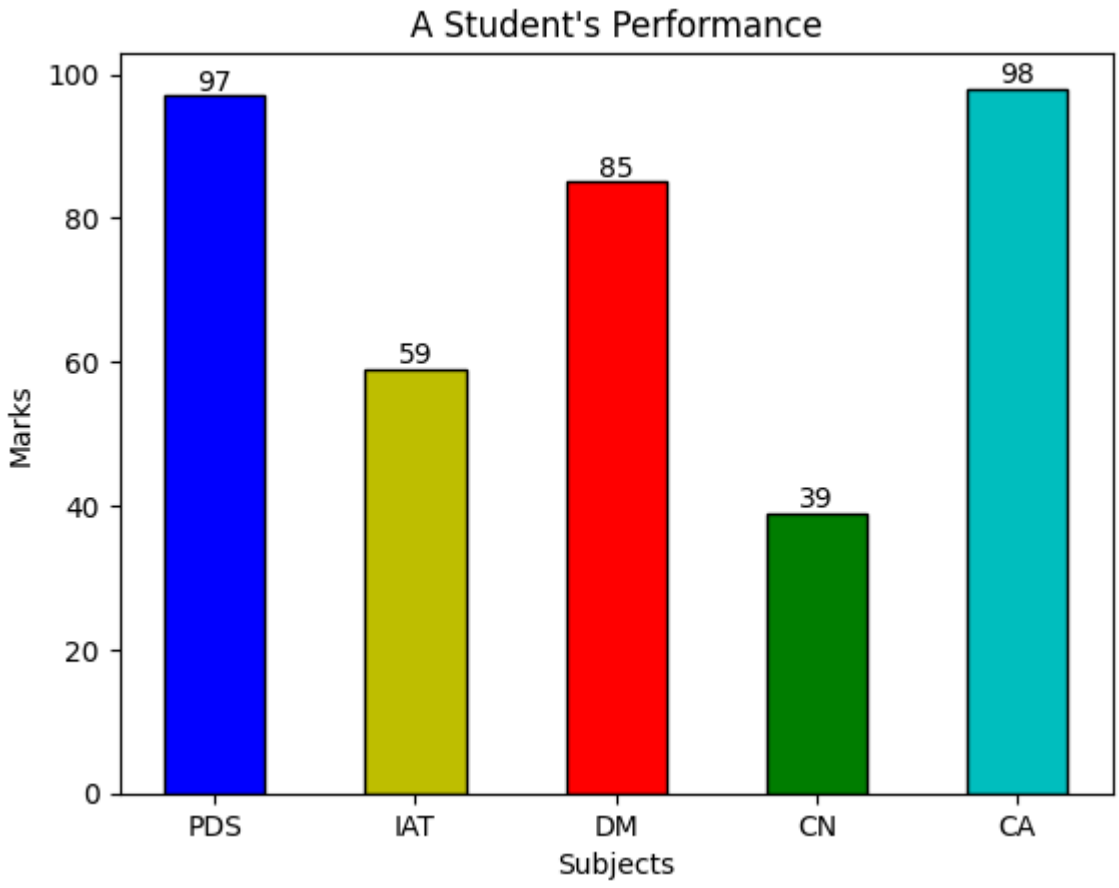
In [29]:
```
subject = ["PDS","IAT","DM","CN","CA"]
```

In [31]:
```
marks = np.random.randint(35,101,5)
```

In [33]:
```
marks
```

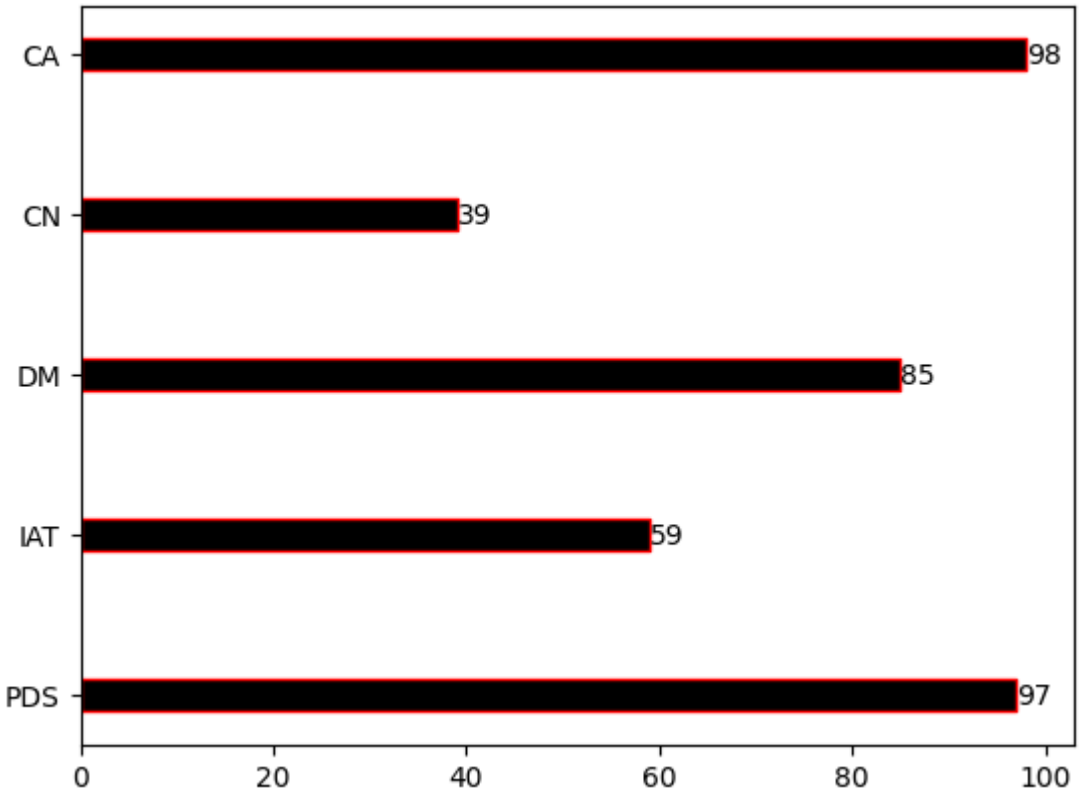
Out[33]:
```
array([97, 59, 85, 39, 98], dtype=int32)
```

In [35]:
```
bars = plt.bar(subject, marks, width = .5, color = ["b","y","r","g","c"],
               edgecolor = "k")
plt.bar_label(bars)
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.title("A Student's Performance")
plt.show()
```



Horizontal Bar Chart

In [37]:
```
bars = plt.barh(subject, marks, color = "k", edgecolor = "r", height=0.2)
plt.bar_label(bars)
plt.show()
```



Scatter Plot

In [40]:
```
study_hours = range(1,11)
```

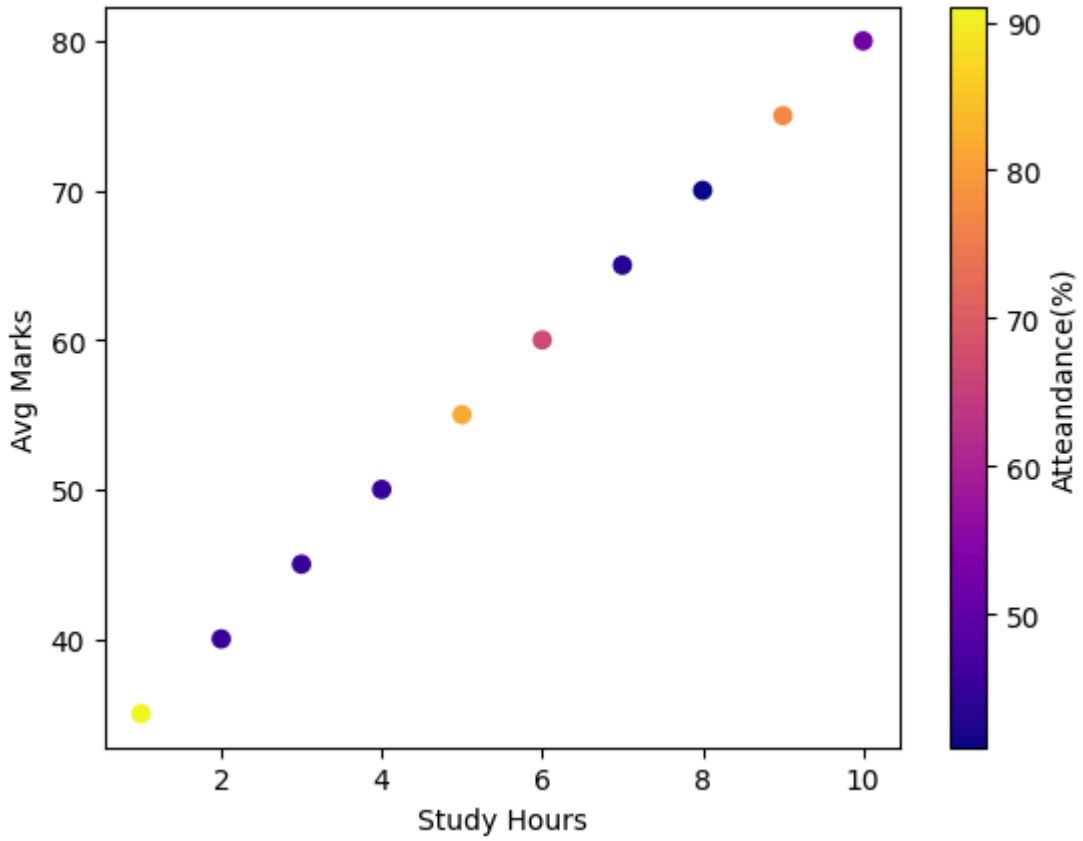
In [42]:
```
marks = range(35,81,5)
```

In [44]:
```
study_hours
```

Out[44]:
```
range(1, 11)
```

In [46]:
```
attendance = np.random.randint(35,100,10)
```

In [48]:
```
plt.scatter(study_hours, marks, c = attendance, cmap = "plasma")
plt.colorbar(label = "Atteandance(%)")
plt.xlabel("Study Hours")
plt.ylabel("Avg Marks")
plt.show()
```



In [50]:
```
attendance = np.random.randint(35,100,10)
```

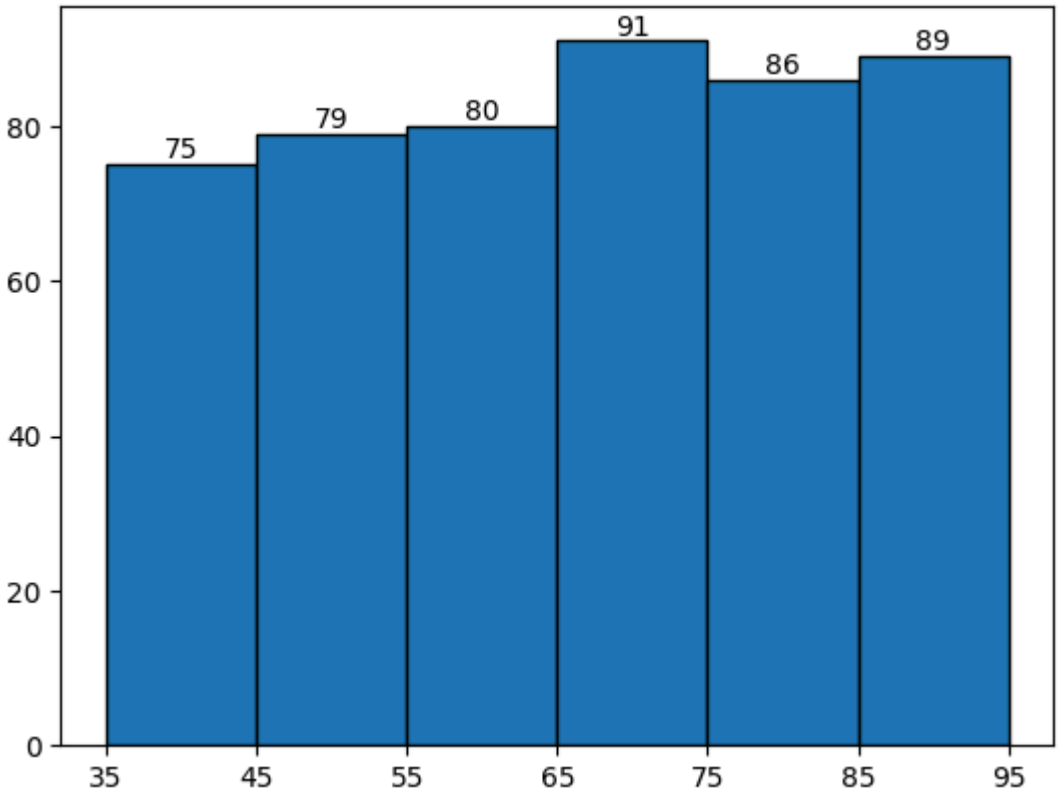
Histogram

In [52]:
```
import numpy as np
```

In [54]:
```
import matplotlib.pyplot as plt
```

In [56]:
```
data  = np.random.randint(35,96,500)
```

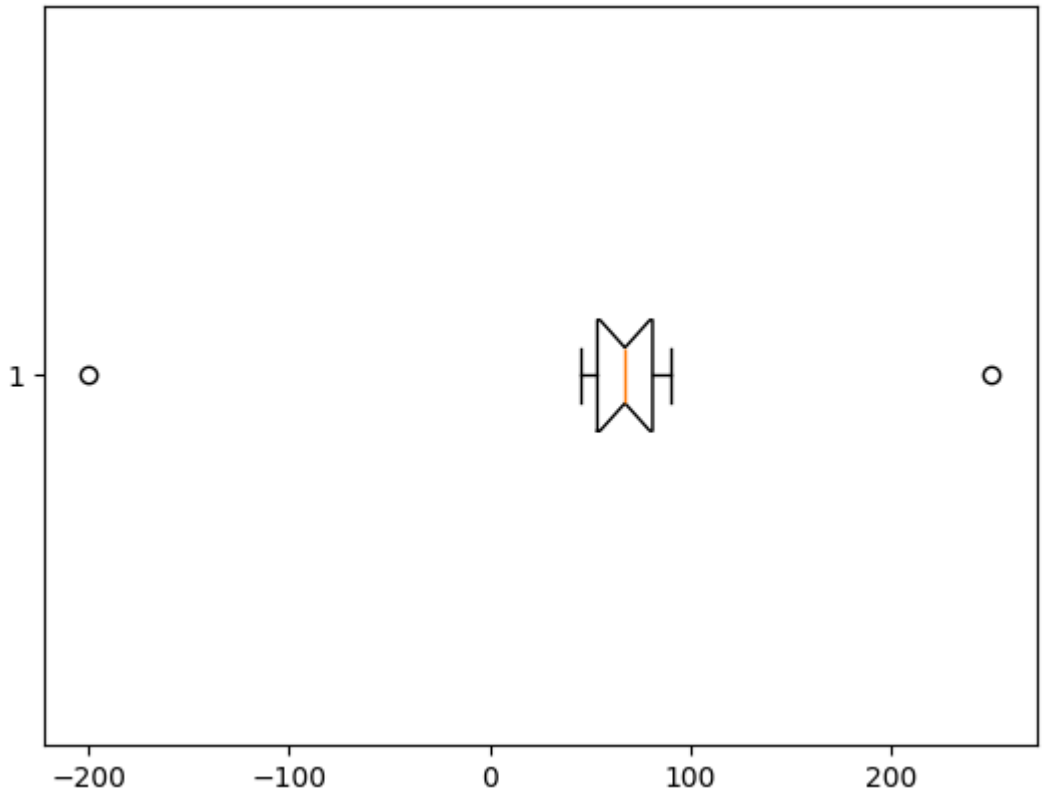
In [58]:
```
x, y, bars = plt.hist(data, bins = [35,45,55,65,75,85,95], edgecolor = "k")
plt.bar_label(bars)
plt.xticks([35,45,55,65,75,85,95])
plt.show()
```



Boxplot

In [60]:
```
data  = [-200,45, 50, 55, 60, 65, 70, 75, 80, 85, 90,250]
```

In [62]:
```
plt.boxplot(data, vert = False, notch= True)
plt.show()
```



In [64]:
```
PDS = np.random.randint(25,101,10)
```

In [66]:
```
DM = np.random.randint(40,101,10)
```

In [68]:
```
IAT = np.random.randint(35,96,10)
```

In [70]:
```
plt.boxplot([PDS, DM, IAT])
# plt.xticks(["PDS","DM","IAT"])
plt.show()
```

